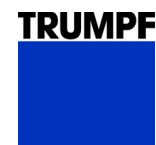


Technical Quotation



S&W Metal Products
200C N Furnace Street
Birdsboro, Pennsylvania 19508
USA

TruBend Cell 5170 (S) (KB36)

Quotation number: 00166103/1 from 06/25/2025
TRUMPF Project ID: 0000701605
valid to: 09/23/2025



All illustrations are examples. The product may differ from the picture.

Regional Sales Manager

Mr James Rogowski

Phone (860) 3845467
E-mail james.rogowski@trumpf.com

Sales Representative

Mr Chris Barsanti

Mobile (610) 256-0711
E-mail CBarsanti@midatlanticmachinery.com

Product Manager

Julian Schorpp

Phone +1 224-288 8455
E-mail julian.schorpp@trumpf.com

Technical Quotation



**S&W Metal Products
200C N Furnace Street
Birdsboro, Pennsylvania 19508
USA**

Dear Customer,

Thank you for your interest in TRUMPF technology. In response to your request we are pleased to provide the following quotation.

TRUMPF machines are designed to enable metal fabricators to compete and win in today's intensely competitive landscape.

It is most important to us that you benefit from your TRUMPF machine each and every day. TRUMPF accessories and services is the key to attain the full potential of your new TRUMPF machine and will ensure a maximum uptime, low operating cost and maintain its high capital value. We bundle these products under the category TruServices. TRUMPF knows it is our obligation to develop and continuously improve TruServices to ensure your competitive advantage for the lifetime of your machine. This is why we include superior services with your new machine such as our TRUMPF warranty, service agreements, consumables, training, installation and data-driven services for monitoring an optimal performance.

As the premier laser manufacturer in the world, and the largest manufacturer of fabrication equipment in the United States, TRUMPF is positioned to offer innovative machines of very high reliability and quality. TRUMPF has been in business for over 100 years and spends approximately 10% of its revenues each year on research and development that result in continued enhancements of existing products along with innovative new technologies to solve tomorrow's fabrication needs.

While our fabrication machines set themselves apart from our competitors' products through their flexibility and productivity, the company also distinguishes itself through dedication to customer training and support. Thousands of students pass through our US customer training center here in Farmington, Connecticut each year, learning to program, operate and maintain their TRUMPF products. We are also constantly optimizing our technical service group, keeping pace with our installed base of machines. Our dedication to continuous innovation and customer support has helped the TRUMPF Group to exceed \$5 billion in sales worldwide.

If you have not already had the opportunity to experience our 500,000 square foot manufacturing facility in Farmington, CT, we would like to offer you a visit. Meeting the people of TRUMPF while getting a firsthand look at our facility, including our new SmartFactory, which is opening early next year, our products and our lean manufacturing operations will open your eyes to the TRUMPF philosophy of providing machine tools and lasers which provide the lowest cost per part. If Connecticut is not your backyard, our technology is also showcased in our SmartFactory near Chicago, IL and in our Technology Center in Costa Mesa, CA. Contact your local TRUMPF sales person to arrange a visit.

Thank you once again for considering TRUMPF.

Burke Doar
Executive Vice President

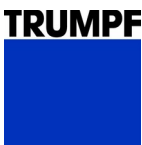


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1. Quotation

1000	TruBend Cell 5170 (S) (KB36)	1 PCE	0.00 USD
	• Machine • Press force: 1700 kN/bending length: 4250 mm • C frame • Press beam • Spherical bearing of the press beam with press beam tilting • On-Demand Servo Drive with 4-cylinder technology • Distance measurement system (Y axis) with dynamic deflection compensation • Hydraulic oil filling • Extension of the open height/stroke • E-Learning Package Technology TruBend • Crowning • CNC-controlled wedge crowning • Backgauge • Gauge fingers (two pieces) • Tool clamp • TRUMPF hydraulic upper tool clamp • TRUMPF hydraulic lower tool clamp • Control • Touchpoint TruBend machine control • User management system • Bending technology data • Data transmission • USB interface • RJ45 network interface • Network capability • Remote support interface • Operation • Operating unit with foot switches • Communications rail • MobileControl		

Technical Quotation



- Safety
 - CE marking and EC Machinery Directive
 - Standard safety device
 - Rear roller gate
- Advanced standard configuration
 - 4-axis backgauge (X, R, Z1 and Z2 axis) including sensor system
 - Pneumatic lower tool displacement (I axis)
 - Integrated TRUMPF technology data
 - Installation, initial start-up and instruction
 - Electrical cabinet air conditioning up to an ambient temperature of 35 °C
 - Right support arm
 - Data use agreement

Backgauge

2000	6-axis backgauge (X1, X2, R1, R2, Z1 and Z2 axis) including sensor system	1 PCE	45,000.00 USD
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Lower tool displacement

3000	CNC-controlled lower tool displacement (I axis)	1 PCE	8,000.00 USD
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Control

Data transmission

4000	Remote Control Interface (RCI) • Connection of third-party systems for remote control of the machine and production plan • Machine messages to MDA/MES software	1 PCE	15,000.00 USD
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Angle assistants

5000	ACB Wireless (Automatically Controlled Bending) • Machine-side preparation for use of sensor tools	1 PCE	0.00 USD
6000	ACB Laser (Automatically Controlled Bending)	1 PCE	35,000.00 USD

Operation

7000	Additional MobileControl	1 PCE	2,000.00 USD
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Technical Quotation



8000	MobileControl Pro	1 PCE	6,000.00 USD
9000	Tool Indicator	1 PCE	7,000.00 USD
10000	Front work area illumination • LED	1 PCE	0.00 USD
11000	Rear work area illumination • LED	1 PCE	0.00 USD
12000	2D code scanner DMC Bluetooth • Wireless (Bluetooth) • Reading of 2D codes from paper or sheet metal parts	1 PCE	4,500.00 USD

Safety

13000	BendGuard Automatic	1 PCE	17,000.00 USD
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ToolMaster

14000	ToolMaster (KB37) Machine	1 PCE	230,000.00 USD
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- Robust machine frame
 - Upper tool magazine with three levels
 - Lower tool magazine with three levels
 - CNC-controlled transfer unit for upper and lower tools
 - Push chain manipulators for upper and lower tools
 - Magnet gripper system
 - CNC-controlled turning station for upper and lower tools
 - Position monitoring
 - Right-hand installation
 - Electrical cabinet air conditioning up to an ambient temperature of 45 °C
- Control
 - Integrated into the Touchpoint machine control
- Safety
 - Light curtain (vertical and horizontal) with fix reflector

Requirement for the ToolMaster (KB37): Machine Preparation for ToolMaster or equivalent functions, or machine without BendGuard

Technical Quotation



- B23: Included in package 3 - Tool Change Upgrade

15000	Storage expansion (fourth level) (KB37)	1 PCE	12,000.00 USD
	• One additional level each as storage for upper tools and lower tools		
16000	Additional cover, transparent (KB37)	1 PCE	8,000.00 USD
	• Plexiglass panes as partial side and rear covering		
17000	Work area illumination (KB37)	1 PCE	3,500.00 USD
	• LED		
18000	Roof cover (KB37)	1 PCE	3,500.00 USD
	Special options		
19000	Special paint finish	1 PCE	20,000.00 USD
	• Maximum two colors		
20000	Electrical cabinet air conditioning up to an ambient temperature of 40 °C	1 PCE	3,000.00 USD
21000	Tank heating	1 PCE	1,000.00 USD
22000	ToolMaster mechanical and electrical interface (KB37)	1 PCE	10,000.00 USD
	Documentation		
23000	TruBend documentation	1 PCE	0.00 USD
24000	BendMaster (150)	1 PCE	859,000.00 USD
	• BendMaster • Electric drives • Double sheet detection • Sheet sensor for contour recognition • Rotary transmission pass-through for simple gripper change • Gripper coupling on the BendMaster • Vacuum generation for vacuum gripper • NC regripping station • Safety fencing with access door		

Technical Quotation



- Stable base body and manipulator

Control

25000	Additional monitor on cell control operating panel • 21.5" monitor	1 PCE	0.00 USD
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Material handling

26000	Gripper changing console BM150 (position according to standard layout)	1 PCE	6,000.00 USD
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27000	Mobile fanning magnet	1 PCE	3,000.00 USD
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Floor track extension

28000	10 m floor track extension	1 PCE	30,000.00 USD
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29000	12 m floor track extension	1 PCE	45,000.00 USD
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30000	14 m floor track extension	1 PCE	60,000.00 USD
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31000	16 m floor track extension	1 PCE	75,000.00 USD
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Handling equipment

32000	Conveyor belt for unloading 5000 × 1000 mm (position according to standard layout) • Maximum two pieces • Number of conveyor belts depends on floor track length	1 PCE	24,000.00 USD
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33000	Pallet conveyor for loading and unloading 5400 × 800 mm (position according to standard layout) • Number of pallet conveyors depends on floor track length	1 PCE	27,500.00 USD
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34000	Chain conveyor for loading and unloading 5725 x 1460 mm (steel pallet 3250 x 1460 mm) • Number of possible chain conveyors depends on floor track length	1 PCE	85,000.00 USD
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35000	Control unit for conveyor belt, pallet conveyor and/or chain conveyor • For maximum 6 x handling equipment	1 PCE	8,000.00 USD
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Technical Quotation



Vacuum gripping technology

36000	Multi-gripper Small - Gripper dimensions adjustable approx. from 140 × 120 mm to 240 × 240 mm 180 degree rotary axis for greater productivity	1 PCE	11,500.00 USD
37000	Multi-gripper Medium Gripper dimensions adjustable approx. from 495 × 310 mm to 1105 × 580 mm	1 PCE	15,500.00 USD
38000	Vacuum gripper kit Material for three grippers	1 PCE	10,500.00 USD
39000	Gripper coupling	1 PCE	2,000.00 USD

Pivoted-jaw gripper technology

40000	Sheet removal station with linear axis including pivoting-jaw gripper (position according to standard layout)	1 PCE	45,000.00 USD
41000	Sheet removal station with rotary table including pivoting-jaw gripper (position according to standard layout)	1 PCE	55,000.00 USD
42000	Additional pallet for sheet removal station	1 PCE	3,000.00 USD
43000	Add. pallet for sheet removal station - rotary table	1 PCE	6,000.00 USD

Other options

44000	Safety fence with transparent panes upgrade - Safety fence with transparent panes instead of safety fence with wire mesh - Required for individual layout requests that require modification of the safety fence.	1 PCE	32,000.00 USD
45000	Left side door for safety fence (position according to layout)	1 PCE	4,000.00 USD
46000	Lift gate including storage interface (position according to standard layout)	1 PCE	20,500.00 USD

Technical Quotation



Modified standard options

47000	Sheet removal station alternative position • Alternative position in cell layout • Engineering (cable ducts, cable sets, layout)	1 PCE	3,000.00 USD
48000	Gripper changing console(s) alternative position • Alternative position in cell layout • Engineering (cable ducts, cable sets, layout)	1 PCE	3,000.00 USD
49000	Lift gate alternative position • Alternative position in cell layout • Engineering (cable ducts, cable sets, layout) • Modified safety fence	1 PCE	5,000.00 USD

Documentation

50000	BendMaster documentation	1 PCE	0.00 USD
51000	Standard TRUMPF TruServices	1 PCE	0.00 USD

Warranty:

Standard 1-year warranty with unlimited hours

Training:

2 seats of Operator Training TruBend 5000 Series
2 seats of Programming Course TruTops Bend Automation TruBend Cell 5000 (online)
2 seats of Operator Course TruBend Cell
4 days application support at customer site

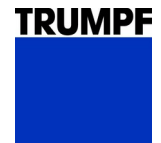
Tooling:

With the purchase of this TruBend press brake we offer a special discount of \$1,500 off the list price of one stand-alone tool cabinet by Apfel. Tool cabinets help to maintain your valuable press brake tooling and keep every tool at the right location in your facility. With a high-quality and ergonomic design, these cabinets reduce your press brake setup times to a minimum.

Transparency Agreement:

Machine will be connected to our Condition Monitoring Platform including the following benefits:
Data-driven diagnostics: Faster and more reliable problem solving based on machine data
Detection and prevention of machine failures and unplanned

Technical Quotation



downtimes

Condition Report via MyTRUMPF: Machine performance data going back 7 days, so that you can analyze them and take the necessary steps to improve your operation
Valid for five years

Augmented Reality Technology:

With ART (Augmented Reality Technology) we'll be able to facilitate your issues, in real time, in the most precise manner available to date. With our complimentary pair of glasses, you can enjoy unlimited use at absolutely no extra cost while your machine is under warranty. After your warranty has expired you can still enjoy unlimited use with the purchase of our Service Agreements.

Installation:

Installation, Start-up and Instruction

Standard Acceptance:

Machine acceptance will consist of producing TRUMPF's standard acceptance part which checks all aspects of machine performance against quoted specifications. This acceptance test will be performed at the customer's location, if requested. Any additional tests must be mutually agreed upon and specified in the purchase order.

52000	Additional Preventive Maintenance visit	1 PCE	6,000.00 USD
53000	Smart Glasses	1 PCE	2,750.00 USD
	Hardware only, but must be purchased with an Agreement		
54000	Additional application support @ Customer	1 PCE	12,000.00 USD
	One week support (incl. travel and lodging costs)		
55000	Sheetmetal Design Class, 3 days @ TRUMPF	1 PCE	4,000.00 USD
	1 seat included, \$3,500 per additional seat		
	In this course, your employees will learn how to streamline the design process and make full use of sheet metal knowledge to design better parts and replace and/or make improvements to current manufacturing		
56000	Oseon Analytics (Complete Package - US Standard) - Valid for new SOFTWARE customer only	1 PCE	8,500.00 USD
	• One user license • Implementation of one TRUMPF machine • Software maintenance for 12 months (for new software customer only) or alignment the software maintenance with the current license(s) (existing software customer)		

Technical Quotation



57000	TruTops Boost Software bundle (Machine w/Automation) - Valid for new TruTops Boost customer only All in-one design and programming solution that works consistently from 3D modeling to NC generation across different technologies. The integrated software offers connected intelligence for the ideal interaction between software and machine. - One TruTops Boost core (floating license) including one Machine Package - Training seat for one student - Software maintenance contract for a 12-month period - Telephone & remote support and software updates	1 PCE	0.00 USD
58000	Additional core (floating license) of TruTops Boost - Existing TruTops Boost license(s) as well as valid software maintenance is required Machine package(s) from the first core included. Current Software maintenance contract required / follows first core validity. Training not included	1 PCE	11,000.00 USD
59000	Data Exchange Package for 3D CAD models - Existing TruTops Boost license(s) as well as valid software maintenance is required Read and write: VDA-FS (VDA), Parasolid (X_T, X_B, XMT_TXT, XMT_BIN) Read: Inventor (IPT, IAM), NX (PRT), Pro/E (PRT, ASM, XPR, XAS), SolidEdge (par, psm, asm), Solidworks (SLDPRT, SLDASM)	1 PCE	2,700.00 USD
60000	Assembly Importer - Existing TruTops Boost license(s) as well as valid software maintenance is required Automated dissolving of assemblies in single parts in all supported file formats and automatic transfer into HomeZone (parts management)	1 PCE	2,700.00 USD
61000	Interface Package Catia V5 - Existing TruTops Boost license(s) as well as valid software maintenance is required Supports direct import and export of the file format CATIA V5 (MODEL, EXP, CATpart, CATproduct, CGR)	1 PCE	4,500.00 USD
62000	TruTops Boost Design Programming (webinar) This course is designed for customers who wish to learn design in TruTops Boost without having to travel to TRUMPF. The goal of the course is to provide students with the core knowledge and skills to successfully draw 2D or 3D parts.	1 PCE	2,400.00 USD
63000	Extended Warranty + 1 PM Mission Warranty extension with 1 preventative maintenance mission per year. (1 year max)	1 PCE	44,000.00 USD

Technical Quotation



64000	Service Agreement Remote	1 PCE	750.00 USD
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- Technical service hotline
- Remote Support: Online Remote diagnostics and troubleshooting
- Technical Guides

Remote Extension 24/7

- Technical service hotline now available around the clock
- Remote support now available around the clock

Visual Assistance

- Recording and exchanging audio, image and video files
- Editing images with real-time transmission
- Chat function for convenient communication

Condition Monitoring: option which identifies anomalies using machine data analysis, includes:

- Proactive expert contact by phone
- Proactive recommendations for action by mail or service app
- Proactive problem solving by TRUMPF service technicians

Condition Monitoring recommended for compatible machines at an additional fee of \$800

65000	Service Agreement Maintenance	1 PCE	13,000.00 USD
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- Technical service hotline
- Remote Support: Online Remote diagnostics and troubleshooting
- Technical Guides
- Maintenance performed to specifications of the maintenance checklist

On-site maintenance

- One preventative maintenance per contract year

Remote Extension 24/7

Technical Quotation



- Technical service hotline now available around the clock
- Remote support now available around the clock

Visual Assistance

- Recording and exchanging audio, image and video files
- Editing images with real-time transmission
- Chat function for convenient communication

Parts discount

- 10% Discount on service parts, long-life consumables and consumables

Condition Monitoring: option which identifies anomalies using machine data analysis, includes:

- Proactive expert contact by phone
- Proactive recommendations for action by mail or service app
- Proactive problem solving by TRUMPF service technicians

66000	Service Agreement Maintenance + Onsite Repair Coverage	1 PCE	30,000.00 USD
	• Technical service hotline • Remote Support: Online Remote diagnostics and troubleshooting • Technical Guides • Maintenance performed to specifications of the maintenance checklist		

On-site maintenance

- One preventative maintenance per contract year

Remote Extension 24/7

- Technical service hotline now available around the clock
- Remote support now available around the clock

Technical Quotation



Visual Assistance

- Recording and exchanging audio, image and video files
- Editing images with real-time transmission
- Chat function for convenient communication

Parts discount

- 10% Discount on service parts, long-life consumables and consumables

Onsite repair coverage

- Labor and travel expenses for repair missions are included

Condition Monitoring: option which identifies anomalies using machine data analysis, includes:

- Proactive expert contact by phone
- Proactive recommendations for action by mail or service app
- Proactive problem solving by TRUMPF service technicians

67000	Service Agreement Performance	1 PCE	44,000.00 USD
	• Technical service hotline • Remote Support: Online Remote diagnostics and troubleshooting • Technical Guides • Maintenance performed to specifications of the maintenance checklist • Repair missions • Service parts for repair missions		

On-site maintenance

- One preventative maintenance per contract year

Remote Extension 24/7

Technical Quotation



- Technical service hotline now available around the clock
- Remote support now available around the clock

Visual Assistance

- Recording and exchanging audio, image and video files
- Editing images with real-time transmission
- Chat function for convenient communication

Parts discount

- 10% Discount on service parts, long-life consumables and consumables

Extended parts coverage

- Coverage of long-life consumables for repair missions

Condition Monitoring: option which identifies anomalies using machine data analysis, includes:

- Proactive expert contact by phone
- Proactive recommendations for action by mail or service app
- Proactive problem solving by TRUMPF service technicians

All freight charges, insurance premiums, duties, tariffs, and taxes are the responsibility of the Buyer.

If a service agreement is included: Service agreement will start after the warranty period of the machine expires.

2. Terms and conditions

Terms of payment

Payment terms:

20% deposit with purchase order

70% ten (10) days after shipment

10% ten (10) days after installation completion

TRUMPF, INC. TERMS AND CONDITIONS OF SALE FOR MACHINE TOOL EQUIPMENT AND AUTOMATION

This Agreement governs the sale of commercial off-the-shelf products from Seller's product lines (hereinafter referred to as "Goods" or "Equipment"), including products sold as a Smart Factory. A Smart Factory connects two or more products together. Seller makes all quotations and accepts orders for such Goods/Equipment only on the terms and conditions stated herein ("Terms of Sale").

1. ORDER ACCEPTANCE:

All orders are subject to acceptance only at Seller's facility in Farmington, Connecticut. These Terms of Sale shall be deemed accepted by Buyer upon Seller's receipt of Purchase Order from Buyer. No condition stated by Buyer shall be binding upon Seller if in conflict with, inconsistent with or in addition to the Terms of Sale, unless expressly accepted in a writing signed by Seller. In the event of conflict or differences in the terms or conditions of Buyer's Purchase Order and the Terms of Sale herein, the Terms of Sale shall govern.

2. PRICES:

All prices are: (a) Seller's current prices and are subject to change without notice at any time prior to acceptance of Buyer's order; (b) subject to all federal, state and local taxes upon the production, sale or shipment of the Goods sold hereunder, now or hereafter becoming effective, and if not included in the invoice, such amount may be invoiced later, and Buyer shall pay all such taxes.

If any new tariffs, duties, or similar governmental charges are imposed, enacted, or materially altered, and such measures increase the cost to Seller of providing the Goods contemplated herein, Seller shall adjust the prices charged to Buyer to reflect such increased costs. Seller shall provide Buyer with written notice of any such cost increase. Buyer agrees to pay for the additional cost, which shall become effective upon receipt of notice.

Technical Quotation



3. SHIPMENTS:

(A) Domestic: All orders are FCA Seller's plant in Farmington, Connecticut (or such warehousing facilities as Seller may establish) Incoterms 2020.

(B) International: All prices are in US Dollars and are, at Seller's election, FCA Farmington CT (or other warehousing facilities as Seller may establish) or DPU Port of Entry Incoterms 2020.

(C) All freight charges, insurance premiums, duties and taxes (including tariffs) are the responsibility of the Buyer.

4. PAYMENT:

All payment terms are subject to Seller's approval of Buyer's credit.

(A) Equipment shipping into US or Canada:

20% deposit with purchase order

70% Net 10 days after shipment

10% Net 10 days from Installation Completion.

Title to the Goods shall pass to the Buyer only upon receipt of final and full payment of the Goods by Supplier, from Buyer.

(B) Equipment shipping into Mexico:

20% deposit with purchase order

60% paid prior to shipment

20% Net 10 days from Installation Completion

Title to the Goods shall pass to the Buyer only upon receipt of final and full payment of the Goods by Supplier, from Buyer.

PLEASE NOTE: During the payment process, TRUMPF will never ask you to transfer to a changed bank account. If you notice anything suspicious, please contact your TRUMPF representative, and in the case of urgent suspicion, contact security@trumpf.com.

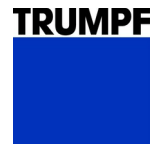
5. DELIVERY:

The scheduled shipment date is an estimate and is subject to filling prior orders and delays caused by strikes, accidents, shortages, acts of civil authority or other causes beyond Seller's control. Buyer's acceptance of delivery time from the shipper shall constitute a waiver of any claim for delay.

6A. Preparation of Installation Site:

Upon acceptance by Seller of any Purchase Order hereunder, Seller shall provide Buyer with a Pre-Installation Manual setting out requirements for site preparation at the installation site. Buyer is responsible for preparing the installation site as set out in the Pre-Installation Manual, for providing all utilities within rated parameters as stated in the Pre-Installation Manual, and for inspecting, rigging and placement of the Goods at the installation site.

Technical Quotation



6B. Inspection Upon Delivery:

Upon arrival of the Goods at the installation site, the Goods shall be inspected for shortages and damage in transit by authorized personnel of both Seller and Buyer. Buyer shall notify Seller in writing of any such claims within ten (10) days after Buyer discovers or should reasonably have discovered facts upon which the claim is based, but in no event more than thirty (30) days after Installation Completion under subsection C below. Failure of the Buyer to give written notice of a claim within the time-period or in the form specified above shall be deemed to be a waiver of such claim. If any repairs are made by unauthorized personnel, Seller reserves the right to withhold warranty support and to charge Buyer per then current time and material rates for services provided if Seller determines that the root cause of a defect requiring service was caused by a third- party vendor supplied product or service.

6C. Acceptance Testing and Acceptance:

Notwithstanding prior inspection or testing, payment, or passage of title, all Goods shall be subject to final acceptance testing ("Final Acceptance Testing") after installation at the installation site to validate performance against Seller's quoted Technical Specifications. Final Acceptance Testing will be performed by Seller's operator and consist of (i) Seller's standard acceptance-test procedure which checks all aspects of Goods performance against Seller's quoted Technical Specification and (ii) any additional tests that are mutually agreed upon and specified in the purchase order or statement of work referencing these Terms and Conditions. Upon successful completion of the Final Acceptance Testing, the Goods shall be accepted, and Buyer shall sign and deliver to Seller the Seller's acceptance certificate ("Installation Completion"). Notwithstanding the foregoing, the Goods including any software shall be deemed accepted and the written acceptance form certifying Installation Completion shall be deemed executed by Buyer upon the earlier of either of the following two conditions; (i) in the event the Goods are placed into production by Buyer at the Installation Site; or (ii) no later than 3 months after the date of delivery as determined by the terms of shipment.

7. EQUIPMENT WARRANTY: Any Equipment sold hereunder which (1) at the installation site, (2) have been properly installed and maintained by authorized persons, and (3) have been operated within the limits of rated and normal usage, are warranted to conform to Seller's quoted Technical Specifications and to be free of defects in material and workmanship, as determined by Seller's inspection, for a period of one year, or any purchased extended period, coming into effect upon installation completion. To further facilitate efficient warranty services if the Goods are sold as a Smart Factory, the warranty will come into effect upon installation completion. If the Buyer places a portion of the Smart Factory Equipment into production, the warranty will cover that Equipment one year after the entire Smart factory's installation is completed but in no event longer than eighteen (18) months from the date the Buyer placed that Equipment into production. The terms of this warranty do not apply to any Goods which have a life, under normal usages, that is inherently less than one year. Product demonstrations, test parts, time studies, production estimates and other such particulars furnished to Buyer are only Seller's estimate and do not create a warranty. Within the warranty period, Seller will repair or replace without cost to Buyer any product or parts covered by the warranty which Seller finds to be defective in material or workmanship, provided that the Buyer gives the Seller prompt notice. This shall be the sole and exclusive remedy of the Buyer under this warranty.

If Buyer licenses Applications Software to be delivered with Equipment at time of installation, at the end of the Applications Software warranty period Buyer shall receive a Software Service Agreement for the balance of the Equipment Warranty period.

To facilitate efficient warranty services delivery, Seller may in its sole discretion deliver to the Installation Site a variety of spare parts such as are reasonably believed to be of potential use in completing the Services (such spare parts, together with those parts removed for replacement during Warranty Services, being collectively "Seller's Property.") Buyer will cooperate with Seller in maintaining Seller's Property in a secure location segregated from Buyer's Equipment and Inventory during Services and in providing access during normal business hours for delivery and pick up. Seller is solely responsible for the packaging and cost of shipment of Seller's Property to and from Buyer's Installation Site. Should Buyer elect to retain any of the unused Spare Parts after close of Warranty Services mission, Buyer will be billed for such Parts at a discount of 5% off Seller's List Price as of the date of retention. Services under this warranty are provided during normal business hours and using standard freight delivery. Services and delivery may be otherwise provided dependent upon availability and will be invoiced at TRUMPF Service rates in accordance with TRUMPF Service policies and practices, which are incorporated by reference herein. Seller shall not be liable under any warranty or other obligation if the alleged defect in the Goods does not exist or the defect or defective performance is caused in whole or in part by Buyer's or any third party's misuse, neglect, modifications or additions, the use of non-TRUMPF spare parts or consumables, unauthorized attempts to repair, or by accident or other hazard outside Seller's control. Use of a non-TRUMPF spare part shall void the warranty if the TRUMPF spare part is available under the warranty free of charge. The warranty described herein is granted only to the original purchaser of the Goods and original purchaser's specified end-user and is non-transferrable.

DISCLAIMER OF OTHER WARRANTIES: THE WARRANTY DESCRIBED IN THIS SECTION 7 IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE EXCLUDED

8. DATA SERVICES:

Seller's Equipment is self-monitoring and software may collect certain equipment data recording performance metrics of the Equipment during Buyer's use of the Equipment which can be transmitted to Seller in real time via Seller's secure Cloud Connect services portal as set forth in the technical documentation of the Equipment. Cloud Connect services are activated during installation.

Buyers of Equipment utilizing Cloud Connect may receive expanded applications capabilities or enhanced Cloud Connect notification and technician scheduling throughout the Equipment warranty period and during any period in which the Buyer maintains a Seller Service Agreement providing for such notification and reporting services. Seller uses the data collected for delivery of Goods and Services under warranty and service agreements, and in product research and development. Buyer agrees that such data may be transmitted, processed, and used by TRUMPF as set forth herein.

Buyer can opt out of active machine monitoring by following the procedures set forth in the Software and Operator Manuals. Should Buyer exercise this opt out option, it may not receive Cloud Connect enhanced services, may experience limitations on machine functions or systems integrations that rely on information processed through Cloud Connect or automated data communications, or may be subject to surcharges and increased costs.

Data collected via Cloud Connect may include metadata, logs or other information generated by the operation of the Equipment or Embedded Software, in automated logs or similar records through the normal operation of the Equipment. In some cases, and within additional services, NC programs, technology tables, environmental, technology and operation data may also be collected. All data is secured with the same level of care Seller provides for its own confidential information, but in no event less than reasonable care. Some features in Seller software may enable collection of data from users of Buyer's applications that access or use the software. If Buyer uses these features to enable data collection in its application, Buyer is solely responsible for compliance with any applicable law, including obtaining any requisite user consent and maintaining a prominent privacy policy that accurately informs users about how it uses, collects, and shares data.

Seller reserves the right to add and remove services from Cloud Connect offerings at Seller's sole discretion. CLOUD CONNECT SERVICES ARE PROVIDED 'AS IS' WITH NO REPRESENTATION OR WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT LIMITATION: 1) ANY WARRANTY CONCERNING THE AVAILABILITY, ACCURACY OR CONTENT OF THE INFORMATION, OR 2) ANY IMPLIED WARRANTY OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

9A. Product Liability:

With respect to all Goods or materials provided under this Agreement that are Seller's standard product or made to Seller's design or specifications, Seller shall, subject to the limitations of liability governing this Agreement, defend and indemnify Buyer against all third-party claims and liability required to be paid by Buyer to the extent arising out of defective materials, workmanship or design by Seller.

9B. Infringement/Malware:

Seller shall engage in commercially reasonable efforts to monitor the Goods supplied hereunder for infringement of third-party intellectual property and the presence of viruses or other malicious code. TRUMPF represents that (i) the Goods are not known to infringe, violate or misappropriate any intellectual property right(s) of third parties; (ii) that TRUMPF has all of the rights, permits, licenses and authority known to be necessary to perform its obligations hereunder; and (iii) the Goods, including any software and the media it is delivered on, have been scanned for and been found to be free from viruses and other malicious code. Seller shall, subject to the limitations of liability governing this Agreement, defend and indemnify Buyer against all third-party claims and liability required to be paid by Buyer to the extent arising out of Seller's negligence or malfeasance in monitoring the Goods as provided herein or in making the foregoing representations.

9C. Workplace Practices:

It is the responsibility of Buyer to know, understand and comply with the work and safety laws and regulations in effect and governing Buyer's use of the Goods provided hereunder. Buyer agrees that after installation of the Goods provided hereunder, the Goods and all persons other than Seller personnel operating or maintaining such Goods will be deemed under Buyer's exclusive control. To the extent governing law requires inspections, reviews, records keeping and/or after-purchase modifications to the Goods, it is the responsibility of the Buyer to arrange for and comply with such requirements and any associated costs are the sole responsibility of the Buyer. Subject to the limitation of liability governing this Agreement, Buyer shall indemnify Seller against all third-party claims and liability required to be paid by Seller to the extent arising out of (a) any modifications, including but not limited to repairs, made in the Goods by or on behalf of Buyer by persons or parties other than Seller, and/or (b) the Buyer's negligent use of the Goods, including but not limited to use in production with any of the Product safety functions disabled, obstructed or circumvented.

9D. Notice; Consent to Settlement:

It is a condition of indemnification or defense under this Section 9 that in the event that any third party asserts a claim or liability with respect to any matter for which a Party is entitled to indemnification hereunder, the indemnified Party gives prompt written notice to the indemnifying Party of such claim or liability. No settlement of an indemnified claim shall require an admission of liability or impose an affirmative obligation on a Party hereto, including an obligation to indemnify, without the written consent of the Party so affected. Consent shall not be unreasonably withheld.

9E. Insurance:

Each Party shall maintain comprehensive liability insurance coverage in an amount not less than one million dollars per occurrence with coverage of the other Party as an additional insured. Upon request, a Party shall provide the other Party with a certificate of insurance evidencing this coverage.

9F. Excluded Limitations:

The above indemnifications shall not be limited by the availability of insurance coverage to the indemnifying party, or by any protection afforded the indemnifying party under the Workers' Compensation Acts, Disability Acts or other employee benefits acts.

10. LIMITATION OF LIABILITY:

NEITHER PARTY'S LIABILITY ON ANY INDEMNIFICATION OR ANY CLAIM OF ANY KIND, UNDER ANY THEORY, AT LAW OR IN EQUITY, INCLUDING NEGLIGENCE OR STRICT LIABILITY, FOR ANY LOSS OR DAMAGE ARISING OUT OF, CONNECTED WITH, OR RESULTING FROM THE PERFORMANCE OR BREACH OF THE TERMS HEREOF, OR FROM THE DESIGN, MANUFACTURE, SALE, DELIVERY, RESALE, INSTALLATION, TECHNICAL DIRECTION OF INSTALLATION, INSPECTION, MODIFICATION, REPAIR, OPERATION OR USE OF ANY SERVICE, GOOD OR PART THEREOF SHALL IN ANY CASE EXCEED THE PRICE ALLOCABLE TO THE SERVICE, GOOD OR PART THEREOF WHICH GIVES RISE TO THE CLAIM. IN NO EVENT SHALL EITHER PARTY HAVE ANY LIABILITY FOR ANY INCIDENTAL, CONSEQUENTIAL OR EXEMPLARY DAMAGES ARISING OUT OF OR IN CONNECTION WITH A BREACH OF THE CONTRACT SALE OR ANY OTHER DUTY OF THAT PARTY WITH RESPECT TO THE GOODS OR SERVICES OR THIS AGREEMENT, INCLUDING, BUT NOT LIMITED TO, DAMAGES FOR LOST PROFITS, LOST SALES OR INJURY TO PERSONS OR PROPERTY EVEN IF THE PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH LOSSES.

11. FORCE MAJEURE:

Except with respect to payment obligations under this Agreement, neither Party shall be liable for or will be considered to be in breach of or default under this contract on account of any delay or failure to perform as a result of any causes or conditions that are beyond the Party's reasonable control and that the Party is unable to overcome through the exercise of commercially reasonable diligence, including but not limited to pandemics, fire, explosion, flood, storm or other acts of God, disruptions in transportation networks, failures of carriers or utilities, war, embargo, strike, riot, or the intervention of any government authority. If any force majeure event occurs, the affected Party shall give prompt written notice to the other party and use commercially reasonable efforts to minimize the impact of the event.

12. LIMITATION OF ACTIONS: No action for breach of any term of this contract of sale or any other duty of Seller with respect to these Goods may be commenced more than one (1) year after Installation Completion.

**13. CHANGES,
CANCELLATION:**

Any changes to the scope of the Goods and Services under this Agreement must be agreed in a writing executed by both Parties. Orders may not be canceled except by written notice received by Seller prior to shipment. A restocking charge of ten percent of the selling price will be applied for the cancellation of standard items. Charges for the cancellation of special items will be based on non-recoverable expenses accruing to the order sustained by Seller plus ten percent of the selling price. Either Party may cancel an existing order without penalty or decline future orders under the Agreement in the event of (a) a change of control of, or (b) a bankruptcy of the other Party.

**14. INTELLECTUAL
PROPERTY RIGHTS:**

The Goods sold hereunder are of Seller's proprietary design offered in configurations and only with such modification as is customarily available to all buyers in the marketplace without significantly altering the Good's, function or processes. The Goods do not incorporate or embody Buyer's intellectual property, or any technology developed by Buyer or specifically for Buyer. Buyer acquires no greater interest in the Goods, or the intellectual property embodied in the Goods than those rights of use, maintenance and resale as customarily accompany the purchase of off-the-shelf tangible goods. All Seller intellectual property including but not limited to patents, trademarks, trade names, trade secrets, and all modifications and derivatives thereof, are and shall remain the sole and exclusive property of Seller.

Any operating software or instruction that is embedded in or incorporated into the hardware or electrical components of the Equipment as an integral part of the Equipment delivered is referred to herein as "Embedded Software." Embedded Software is licensed, not sold. Seller grants Buyer a nonexclusive, perpetual, fully paid, royalty-free restricted license to Embedded Software, in executable form, solely as embedded as part of or through the Equipment and only in conjunction with Buyer's operation of the Equipment and Services as provided by Seller. Buyer may not install, access, configure or use any Embedded Software (including any APIs, tools, databases or other components of any Embedded Software) separately or independently of the rest of the Equipment, whether for production, technical support or any other purposes, or otherwise attempt to gain direct access to any Embedded Software components, or permit anyone else (including Buyer's customers) to do any of these things. Embedded Software excludes applications softwares formulated to interface with end users of the goods to perform specific business or technical functions ("Applications Software") and which are subject to the TRUMPF End User License Agreement.

15. DESIGN CHANGES:

Acceptance of Purchase Order by Seller determines product version for delivery. The designs and specification of all Goods sold may be subject to subsequent change by Seller without notice and, in the event of any such changes, Seller will have no obligation whatsoever to make similar changes in Goods previously ordered.

**16. RESERVATION OF
SECURITY INTEREST; RISK
OF LOSS:**

Seller hereby reserves, and Buyer hereby grants a security interest in the Goods (and the proceeds thereof) as security for the payment of the unpaid balance of the purchase price and Buyer's performance of its other obligations hereunder. Buyer will execute and deliver to Seller such financing statements as Seller shall request to perfect such security interest. The security interest hereunder shall terminate upon Buyer's payment in full of the purchase price of Goods and Services, and the respective taxes.

The risk of loss passes to Buyer upon delivery of the Goods to the carrier. Insurance against loss or damage to the Goods during shipment is the responsibility of Buyer. Until the entire purchase price for the Goods is paid in full, Buyer will keep the Goods insured against loss or damage by fire and other risks and hazards included with so-called "extended coverage" insurance, in an amount at least equal to such purchase price. Losses under such insurance shall be made payable to Seller and any payments under such insurance shall be paid to Seller and applied to the unpaid balance of the purchase price. Upon request, Buyer will furnish Seller with copies of the policies of such insurance and each renewal thereof.

**17. NONPAYMENT OF
PURCHASE PRICE:
STORAGE AND
COLLECTION COSTS:**

If Buyer shall not pay the full purchase price within 30 days from the date of Installation Completion of the Goods; Buyer will pay Seller thereafter an additional one and one-half percent (1-1/2%) per month on the unpaid balance of the purchase price until paid in full. Such charge shall be added to and become an additional part of the purchase price for the Goods. Buyer also will pay all storage costs for the Goods after the scheduled delivery date as well as all costs of collection incurred by Seller in collecting the purchase price for the Goods and enforcing its security interest in the Goods, including, without limitation, reasonable attorneys' fees and expenses incurred by Seller.

18. GOVERNING LAW:

The validity, interpretation and performance of this contract for sale shall be governed by the laws of the State of Connecticut, excluding its conflict of law rules. The United Nations Convention on Contracts for the International Sale of Goods is expressly disclaimed by the Parties with respect to this Agreement and the transactions contemplated hereby.

19. ENTIRE AGREEMENT:

This Agreement constitutes the entire agreement between the Parties pertaining to the subject matter hereof, and supersedes all prior agreements, understandings, negotiations and discussions, whether oral or written, of the Parties pertaining to the subject matter hereof.

Effective Date:

Y2024-07-01

1000 TruBend Cell 5170 (S) (KB36)

Ursprungsland

The machine plant is manufactured by TRUMPF Maschinen Austria GmbH + Co. KG, A-Pasching. Its customs origin is Austria.

Export

Export/shipment of the products to be supplied by TRUMPF in the case of further direct delivery to countries outside the European Community may be carried out under certain conditions (product, destination country, end user, etc.) in accordance EC dual-use ordinance, the German Foreign Trade and Payments Act (AWG, AVW), the Austrian Foreign Trade and Payments Act (AußWG 2011) as well as the US re-export regulations (EAR). If need be, please contact your responsible contact person at TRUMPF – should you not have already been so advised. In the event of an export permit requirement under the Foreign Trade and Payments Act / Dual Use Ordinance / US re-export law, delivery abroad is subject to the approval of the responsible authorities BAFA (Federal Office of Economics, Germany); the Federal Ministry of Science, Research and Industry (BMWi), Austria and, if applicable, the Bureau of Industry and Security (BIS) of the USA. Regarding domestic deliveries to exporting companies, we would like to point out that these companies themselves are responsible for applying for the above export license. The dispatch of products subject to issue of an authorization by TRUMPF (= consignor) on behalf of exporting companies (= exporters) can only take place if the exporter submits a copy of the necessary authorization to TRUMPF. The purchaser undertakes to make available all information and documents (for example, the final declaration of an export permit, etc.) to TRUMPF that are requested by TRUMPF should this be necessary. In the case of re-export of US products or of US products installed in our machines/plants, a re-export authorization may also be required from the responsible US authority, the Bureau of Industry and Security (BIS). If need be, please contact the contact person responsible for you at TRUMPF.

24000 BendMaster (150)

Export

Export delivery/transfer/delivery within Germany of the products to be supplied by TRUMPF in the case of further direct delivery to countries outside the European Community may be carried out under certain conditions (product, destination country, end user, etc.) in accordance with the German Foreign Trade Law, the EC Dual Use Regulation, the Swiss Goods Control Act (GKG) or the US re-export regulations (EAR). If need be, please contact your responsible contact person at TRUMPF – should you not have already been so advised. If subject to the issue of an export permit according to the Foreign Economic Law / Dual Use Regulation / Goods Control Law / US Re-export Law, the delivery abroad is subject to approval by BAFA (Bundesamt für Wirtschaft/Federal Office of Economics in Eschborn) responsible for Germany; the SECO (Secrétariat d'Etat à l'économie), or the Bureau of Industry and Security (BIS) of the USA (restrictions under US law, applicable for the re-export of US goods). Regarding domestic deliveries to exporting companies, we would like to point out that these companies themselves are responsible for applying for the above export license. The dispatch of products subject to issue of an authorization by TRUMPF (= consignor) on behalf of exporting companies (= exporters) can only take place if the exporter submits a copy of the necessary authorization to TRUMPF. The purchaser undertakes to make available all information and documents (for example, the final declaration of an export permit, etc.) to TRUMPF that are requested by TRUMPF should this be necessary.

Ursprungsland

Automatisierung - Pasching

The automation is manufactured by TRUMPF Maschinen Austria GmbH + Co. KG, A-Pasching. Its customs origin is Austria.

64000 Service Agreement Remote

Contract term

Automatically renewing after the end of the term of one year

3. Additional information about the order range

1000 TruBend Cell 5170 (S) (KB36)

Machine

C frame

- The robust machine frame consists of a welded construction: a press table, two side columns and connecting beams
- The entire machine design was calculated using the finite element method (FEM) and designed to resist fatigue
- The stable frame design ensures precise results even after years of continuous operation under maximum loads

Press beam

- The press beam is characterized by high rigidity and therefore ensures exact bending results
- The press beam is guided using doubled roller bearings at each side
- Guide rollers in the Z direction stabilize the press beam and prevent radial forces on the cylinder seals
- High, constant bending accuracy is achieved using hardened bars and the adjustable "zero-play" setting

Spherical bearing of the press beam with press beam tilting

- Spherical connection between press beam and cylinder
- The tilt position enables the production of geometric shapes such as funnels and allows simple corrections to be made when producing long profiles
- The spherical bearing prevents cylinder damage (leakage) due to radial forces caused by press beam tilting
- Refer to the technical data for the maximum press beam tilt

On-Demand Servo Drive with 4-cylinder technology

- Very quiet and highly energy-efficient electro-hydraulic drive
- Position control is performed by two synchronous servo motors which only operate during the bending process
- The high axis speeds and accelerations (rapid downward movement, Y return and closing movement) enable high productivity
- The 4-cylinder technology reduces the deflection of the press beam and thus ensures higher part precision
- The flat design of the 4-cylinder technology increases the bending space in front of the machine

Distance measurement system (Y axis) with dynamic deflection compensation

- The high positioning accuracy of the press beam (Y axis) is ensured by high-precision glass scales
- Despite the robust design of the machine frame, deflection occurs with high press forces. This deflection is measured using precision scales and automatically compensated
- The deflection compensation ensures precise results with varying loads

Hydraulic oil filling (life-long filling)

- The machine is delivered with a full oil tank
- This high-quality oil has a service life of ten years, making oil changes under normal conditions unnecessary

Extension of the open height/stroke

- Usable open height extended by 230 mm to 615 mm
- Y stroke extended by 230 mm to 445 mm
- Significant increase in part variety, especially for possible box depths
- B23: Included in package 1 - Performance Upgrade (already included with TruBend 5170/5230/5320)

Crowning

CNC-controlled wedge crowning

- The crowning value calculated by the control and the necessary adjustment is performed by a servo-spindle motor
- The crowning motor is integrated in the machine table and therefore does not cause an obstruction
- Productivity increase especially for bending at multiple stations or with small lot sizes

Backgauge

Gauge fingers (two pieces)

- Wear-resistant gauge fingers ensure constant flange lengths
- The workpiece can butt against the backgauges at three different positions
- Easier back stop positioning of larger workpieces is possible by resting them on the gauge fingers

Tool clamp

TRUMPF hydraulic lower tool clamp

- Short setup time of the lower tools at the push of a button due to hydraulic clamping
- The tools are clamped with clamp pistons. This guarantees secure clamping of various tool segments
- The tool clamp is hardened

TRUMPF hydraulic upper tool clamp

- Short setup times of the upper tools at the push of a button due to hydraulic clamping
- The tools are clamped with clamp pistons. This guarantees secure clamping of various tool segments
- The tool clamp is hardened
- Requirement for ToolMaster

Control

Touchpoint TruBend machine control

- Combines the advantages of modern multi-touch technology with those of industrial control to create an operating concept with revolutionary simplicity
- The Touchpoint surface is consistently oriented towards requirements of everyday production
- Its intuitive operation resembles that of a tablet or smartphone and it can also be operated efficiently with gloves
- The views are presented so as to focus on the essential matters at hand
- Realistic 3D visualization
- Programs produced using TruTops Bend or TecZone Bend can be opened easily on the control
- 21.5" touchscreen

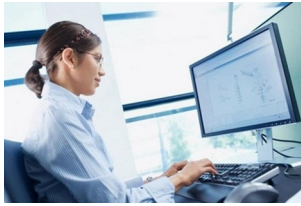
User management system

- Log on to the machine control quickly without entering a password
- Unlocking of machine control using ID card systems
- Different user levels can be configured (standard/expert/customer administration)
- Five ID cards included in the scope of delivery

Data transmission

RJ45 network interface

- The machine control is equipped with an RJ45 network interface (10 or 100 MBit/s) as standard



Remote support interface

- Contact with Technical Service directly via broadband internet connection
- Option of fast reaction
- Minimization of machine downtimes
- High data security due to passwords

Operation

Operating unit with foot switches

- Easy positioning thanks to the compact design and low weight
- Functions: Stroke triggering with EMERGENCY STOP, press beam UP and an EMERGENCY STOP push-button

Communications rail

- The communications rail enables the use of ACB modules, MobileControl and MobileControl Pro
- Up to eight modules (ACB modules/ MobileControl/MobileControl Pro) can be used at the same time in the communications rail

MobileControl

- MobileControl contains the most important control functions such as open/close tool clamps or program start/stop
- MobileControl is placed anywhere in the communications rail and reduces the walking distance to the operating panel

Safety

CE marking and EC Machinery Directive

- Delivery with EU declaration of conformity and CE marking
- Machine is subject to the EC Machinery Directive 2006/42/EC, the EMC Directive 2014/30/EU and further directives where relevant
- In individual cases, the machine can be sold only as "partially completed machinery" with declaration of incorporation. Such a machine must be completed by an integrator.



Rear roller gate

- The roller gate at the rear of the machine is electrically operated and makes the entire machine length easily accessible without obstruction

Advanced standard configuration

4-axis backgauge (X, R, Z1 and Z2 axis) including sensor system

- Additional equipment of the rear stop fingers with one sensor each
- Sensors check the correct position of the sheet on the gauge finger
- Sensor system covers the stop levels 0 and 1

Pneumatic lower tool displacement (I axis)

- The lower tool clamp is mounted on the press table in such a way as to make a displacement in the X direction possible
- Displacement is performed by pneumatic cylinders and both end positions of the I axis are defined by using different spacers
- This function increases productivity and flexibility and at the same time enables easier removal of very high boxes
- Typical applications are quick folding on a tool station, the use of multi V dies or station bending with tools of differing heights (e.g. Z bends or special tools)

Integrated TRUMPF technology data

- The control calculates all bending parameters based on the TRUMPF technology data for standard tools and standard materials
- Modifications for customer-specific materials or tools can be performed by the user

Installation, initial start-up and instruction

- Excluding orders of lifting equipment such as cranes

Electrical cabinet air conditioning up to an ambient temperature of 35 °C

- Cooling unit to prevent overheating of the electrical cabinet
- The electrical cabinet air conditioning guarantees reliable machine operation at ambient temperatures of up to 40 °C

Right support arm

- Connection of control: right-hand side of the machine

Data use agreement

- The machine is delivered with the data usage agreement (standard scope of supply) and offers added value through data transmission in hardware, software and services.
- With the data exchange, you also ensure long-term partnership success:
 - Added value includes higher machine availability and optimized machining processes
 - Ongoing product improvements and other benefits
- Data use agreements are subject to the "Agreement concerning the transfer and use of data", at: www.trumpf.com/s/TCadLT
- With your order, you accept the "Agreement concerning the transfer and use of data".
- If you do not wish to accept the "Agreement concerning the transfer and use of data", please state this explicitly in your order.

2000 6-axis backgauge (X1, X2, R1, R2, Z1 and Z2 axis) including sensor system

- Additional equipment of the rear stop fingers with one sensor each
- Sensors check the correct position of the sheet on the gauge finger
- Sensor system covers the stop levels 0, 1 and 2

3000 CNC-controlled lower tool displacement (I axis)

- Displacement of the lower tool clamp in the X direction is performed by servo-spindle motors
- Maximum flexibility due to freely programmable positions
- Setup times are eliminated as no spacers need to be replaced
- No compressed air connection necessary

4000 Remote Control Interface (RCI)

- Remote control of the machine via an external control system
- Communication between machine and control system via network connection

5000 ACB Wireless (Automatically Controlled Bending)

- Cost-effective bending with high angular accuracy and consistently high quality starting with the first part
- In addition to angle measurement, the ACB Wireless performs automatic regulation to the required set value
- Time-consuming run-in procedures are no longer necessary
- Automatic compensation for changes in material properties
- Measurement is performed by ACB Wireless sensor tools in which two sensor disks with different diameters are integrated
- Communication of the ACB Wireless sensor tools and the control is done via a data rail integrated in the hydraulic upper tool clamp
- Requirement: machine-side preparation is a requirement for the use of ACB Wireless sensor tools
- Notice: ACB Wireless sensor tools and the ACB license are not included in this option
- TruBend 5000 (B23): Availability: 90%
- TruBend Cell 5000 (KB36): Manual mode availability: 90%; automatic mode 80%
- TruBend 8000 (B36): Availability: 90%
- B23: Included in package 5 - Precision Upgrade

6000 ACB Laser (Automatically Controlled Bending)

- Cost-effective bending with high angular accuracy and consistently high quality starting with the first part
- ACB Laser: Beside the angle measurement, automatic regulation to the required nominal value
- Complex run-in procedures are no longer required: system automatically compensates for changes in the material properties
- The measurement is performed optically from two sides using measuring units
- Positioning of measuring units: automatic, CNC-controlled
- The measuring units can be parked at the side outside of the working area (left and right)
- TruBend 5000 (B23): Availability: 90%
- TruBend Cell 5000 (KB36): Manual mode availability: 90%; automatic mode 80%
- B23: Included in package 5 - Precision Upgrade

7000 Additional MobileControl

- MobileControl contains the most important control functions such as open/close tool clamps or program start/stop
- Any positioning of the MobileControl in the communications rail, thereby reducing the walking distances to the operating panel

8000 MobileControl Pro

- Enables the entry of important parameters directly at the machine (e.g. bend forward/back, adjustment values)
- Quick navigation and input with turn-press button
- Entries are transmitted to the control
- Consists of six keys, turn-press button and a 3.2" color display
- MobileControl Pro is placed anywhere in the communications rail and reduces the walking distance to the control
- B23: Included in package 4 - Usability Upgrade

9000 Tool Indicator

- An LED strip is integrated in the upper tool clamp
- Optical display of the positions that the tool stations must be set up in
- Optical display of the tool station at which the next bending operation is to be performed
- Reduction of set up and bending times
- The TRUMPF hydraulic upper tool clamp is required
- B23: Included in package 1 - Performance Upgrade

10000 Front work area illumination

- Very good illumination of the working area in front of the machine
- The lighting is integrated in the machine housing and as such does not present an obstruction
- The modern LED technology works in an energy-saving way, without producing heat and has a very long service life
- B23: Included in package 1 - Performance Upgrade

11000 Rear work area illumination

- Very good illumination of the machine interior
- The lighting is integrated in the machine housing and as such does not present an obstruction
- The modern LED technology works in an energy-saving way, without producing heat and has a very long service life
- B23: Included in package 1 - Performance Upgrade

12000 2D code scanner DMC Bluetooth

- The 2D code scanner enables reading of 2D codes from paper or sheet metal parts for automatic loading of programs
- Entry and search work are reduced and the loading of the wrong programs is eliminated
- Compatible with bar code and Data Matrix Code
- The scanner is wireless, communication using Bluetooth

13000 BendGuard Automatic

- The BendGuard Automatic is an optoelectronic safety device which allows quick and safe bending
- The maximum Y rapid speed is permitted without restriction with foot operation alone
- Automatic setting of the tool height by CNC-controlled height adjustment, without operator intervention
- The BendGuard Automatic can be pushed easily out of the working area for lateral tool setup operations and does not need to be adjusted again afterwards
- The BendGuard Automatic is not sensitive when it comes to bumps and impact (misalignment) because its electronics are separated from its housing
- B23: Included in package 1 - Performance Upgrade

14000 ToolMaster (KB37)

Machine

- Robust machine frame
 - Upper tool magazine with three levels
 - Lower tool magazine with three levels
 - CNC-controlled transfer unit for upper and lower tools
 - Push chain manipulators for upper and lower tools
 - Magnet gripper system
 - CNC-controlled turning station for upper and lower tools
 - Position monitoring
 - Right-hand installation
 - Electrical cabinet air conditioning up to an ambient temperature of 45 °C
- Control
 - Integrated into the Touchpoint machine control
- Safety
 - Light curtain (vertical and horizontal) with fix reflector

Requirement for the ToolMaster (KB37): Machine Preparation for ToolMaster or equivalent functions, or machine without BendGuard

- B23: Included in package 3 - Tool Change Upgrade

15000 Storage expansion (fourth level) (KB37)

- Extension of storage capacity to include a fourth level as additional storage for upper tools and lower tools
- B23: Included in package 3 - Tool Change Upgrade

16000 Additional cover, transparent (KB37)

- Plexiglass panes as partial side and rear covering for the ToolMaster (KB37)

17000 Work area illumination (KB37)

- Very good illumination of the interior space
- The modern LED technology works in an energy-saving way, without producing heat and has a very long service life
- B23: Included in package 3 - Tool Change Upgrade

18000 Roof cover (KB37)

- For dust reduction in the interior space
- B23: Included in package 3 - Tool Change Upgrade

20000 Electrical cabinet air conditioning up to an ambient temperature of 40 °C

- Extended cooling unit to prevent overheating of the electrical cabinet
- The stronger electrical cabinet air conditioning guarantees reliable machine operation at ambient temperatures of up to 40 °C

21000 Tank heating

- The tank heating prevents the oil temperature from dropping too low and thus negatively influencing the travel behavior of the machine
- Tank heating is applicable for machine operation in very cold regions
- B23: Included in package 1 - Performance Upgrade

22000 ToolMaster mechanical and electrical interface (KB37)

- Contains all machine-side interfaces (mechanical and electrical) for connection of the ToolMaster (KB37)
- B23: Included in package 2 - Tool Change Ready

24000 BendMaster (150)

BendMaster

Electric drives

- Electric servo motors with gearbox

Double sheet detection

- Loading of two or more blanks is detected automatically
- Automatic introduction of sheet separation measures

Sheet sensor for contour recognition

- Detects the contour of the sheet and the height of the sheet stack
- Alignment of the component to be produced (plausibility check) through comparison with the contour stored in TecZone Bend
- Use of centering station not needed thanks to pick-up of blank in the correct position

Rotary transmission pass-through for simple gripper change

- Allows internal routing of pneumatic and power cables

Gripper coupling on the BendMaster

- The gripper is connected to the gripper arm by a pneumatic coupling

Vacuum generation for vacuum gripper

- Integrated between B and C axis to save space
- Generates the vacuum for the vacuum gripper

NC regripping station

- Enables the sheet to be set down and then picked up again from the other side
- Integrated in TecZone Bend

Safety fencing with access door

- Entire work area and danger zone closed off with safety fence
- Additional protection with EMERGENCY STOP push-buttons
- As installed by TRUMPF, the system is CE-compliant

Stable base body and manipulator

- Light and sturdy cast aluminum design

25000 Additional monitor on cell control operating panel

- Second monitor on cell control operating panel
- For visualization and display of further documents (e.g. PDF drawings, images, etc.)/ bending machine control
- To support paperless production

26000 Gripper changing console BM150 (position according to standard layout)

- Required for automatic gripper change
- Offers space for three standard or pivoting-jaw grippers
- Use of several gripper changing consoles possible
- Position in the bending cell according to standard layout

27000 Mobile fanning magnet

- Sheet separation possible via cylinder on gripper or special travel motion of BendMaster
- Additional fanning magnet helpful for e.g. highly oiled sheets
- Simple placement on loading stack and freely positionable on the pallet

28000 10 m floor track extension

- Extension of floor track by 2000 mm (= traverse path)
- Enables increase of working area and reaching of deposit position next to the machine
- Series connection of several segments enables continuous extension of path
- Note: As of a machine length of 3 m, extension of the track is recommended depending on the part geometry

29000 12 m floor track extension

- Extension of floor track by 4000 mm (= traverse path)
- Enables increase of working area and reaching of deposit position next to the machine
- Series connection of several segments enables continuous extension of path
- Note: As of a machine length of 3 m, extension of the track is recommended depending on the part geometry

30000 14 m floor track extension

- Extension of floor track by 6000 mm (= traverse path)
- Enables increase of working area and reaching of deposit position next to the machine
- Series connection of several segments enables continuous extension of path
- Note: As of a machine length of 3 m, extension of the track is recommended depending on the part geometry

31000 16 m floor track extension

- Extension of floor track by 8000 mm (= traverse path)
- Enables increase of working area and reaching of deposit position next to the machine
- Series connection of several segments enables continuous extension of path
- Note: As of a machine length of 3 m, extension of the track is recommended depending on the part geometry

32000 Conveyor belt for unloading 5000 × 1000 mm (position according to standard layout)

- Removal of e.g. components which are difficult to stack
- Conveyor belt closed off by a tunnel
- Maximum stack height on conveyor belt 500 mm
- Position in the bending cell according to standard layout

33000 Pallet conveyor for loading and unloading 5400 × 800 mm (position according to standard layout)

- Enables parallel to production insertion and removal of europallets in bending cell
- Parallel to production transport of blanks or bending parts possible
- Closed off by tunnel and light barrier
- The maximum stack height (component-dependent) is 800 mm
- In the standard version, only 2 pallet conveyors can be selected
- Position in the bending cell according to standard layout

34000 Chain conveyor for loading and unloading 5725 x 1460 mm (steel pallet 3250 x 1460 mm)

- Bending cell supplied with new blanks and finished parts removed from the bending cell parallel to production
- The chain conveyor is suitable for the loading and unloading of large pallets with dimensions of 3000 mm x 1460 mm
- Position in the bending cell according to standard layout

35000 Control unit for conveyor belt, pallet conveyor and/or chain conveyor

- Regardless of the selected amount of handling equipment (conveyor belts, pallet conveyors, chain conveyors) only one control unit is required.

36000 Multi-gripper Small

- Suction cup gripper with 4 adjustable arms and a 180° rotary unit
- Gripper dimensions adjustable approx. from 140 × 120 mm to 240 × 240 mm

37000 Multi-gripper Medium

- Suction cup gripper with 8 adjustable arms and individual suction cup activation
- Gripper dimensions adjustable approx. from 495 × 310 mm to 1105 × 580 mm
- Only in combination with at least one gripper changing console

38000 Vacuum gripper kit

- Kit contains several components for assembling three grippers
- Included: profiles, brackets, suction cups, adapters and pneumatic components

39000 Gripper coupling

- Enables customer gripper builds. Required once for each gripper

40000 Sheet removal station with linear axis including pivoting-jaw gripper (position according to standard layout)

- Position components parallel to production from the sheet removal station on a defined transfer point
- Due to integration of a linear axis, two stacks, each with a height of 400 mm, can be processed
- Manipulation of small bending parts using pivoting-jaw gripper
- Less regripping thanks to pivoted-jaw gripper with traversing axis
- Position in the bending cell according to standard layout

41000 Sheet removal station with rotary table including pivoting-jaw gripper (position according to standard layout)

- Integration of the rotary table enables processing of four stacks each with 400 mm height
- Manipulation of small bending parts using pivoting-jaw gripper
- Less regripping thanks to pivoted-jaw gripper with traversing axis
- Position in the bending cell according to standard layout

42000 Additional pallet for sheet removal station

- Minimization of changeover time by loading pallet when cutting blanks
- Small parts pallet is supplied with stop plates and magnets for one stack

43000 Add. pallet for sheet removal station - rotary table

- Supplied with stop plates and magnets for four stacks

44000 Safety fence with transparent panes upgrade

- Safety fence with transparent panes instead of safety fence with wire mesh
- Required for individual layout requests that require modification of the safety fence.

45000 Left side door for safety fence (position according to layout)

- Enables removal of bending parts which are deposited next to the bending machine
- Position in the bending cell according to standard layout

46000 Lift gate including storage interface (position according to standard layout)

- Enables secure connection of bending cell to a storage system
- Opening height is 1200 mm
- Automatic connection to a STOPA store with TC-Cell control
- Usable working area of sheet sensor on the storage pallet is 1500 mm in X direction
- Position in the bending cell according to standard layout

47000 Sheet removal station alternative position

- Alternative, freely selectable position in cell layout
- Engineering (cable ducts, cable sets, layout)

48000 Gripper changing console(s) alternative position

- Alternative, freely selectable position in cell layout
- Engineering (cable ducts, cable sets, layout)

49000 Lift gate alternative position

- Alternative, freely selectable position in cell layout to the left of bending machine
- Engineering (cable ducts, cable sets, layout)
- Modified safety fence

4. Additional information on technical data

TruBend Cell 5170 (S) (KB36)

Machine

Press force	1700 kN
Bending length	4250 mm
Width between columns	3680 mm
Usable open height	615 mm
Press beam tilt	± 10 mm
Throat depth	420 mm
Table width	180 mm
Working height (maximum)	43.1 in - 43.9 in

Speeds

Rapid downward movement	220 mm/s
Max. closing movement (optional) Cell 5000 ¹	10 mm/s
Y return	220 mm/s
X axis for 4-axis backgauge	1000 mm/s
X axis for 6-axis backgauge (optional)	1750 mm/s
R axis	330 mm/s
Z axis for 4-axis backgauge	2500 mm/s
Z axis for 6-axis backgauge (optional)	1750 mm/s

Positioning accuracy

R axis	0.08 mm
Max. axis speed Y axis	0.005 mm
Axis range X axis	0.04 mm

Traverse paths, stop range

Stroke, Y axis	445 mm
Traverse path of X axis	600 mm
Max. stop range in the X direction	860 mm
Traverse path of the R axis	250 mm

Control, hardware

Components	Intel Core i5, 2.7 GHz, Quad Core processor; Windows Embedded Standard 10; 21.5" TFT color display with LED backlight; 1920 x 1080 pixel display resolution (Full HD)
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Technical Quotation



Connected loads

Connected load (IEC) approximately	17.5 kVA
Compressed air inlet port - +/-	6 bar $\pm$ 1 bar

Dimensions and weights

High safety fence	3000 mm
Weight, basic machine	17850 kg
TruBend machine (H)	4970 mm x 1900 mm x 3000 mm

Paint

Textured paint	white = NCS S 0505 R80B, blue = approximately RAL design 250 20 20, black = RAL 9005, silver elements in RAL 9006
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BendMaster (150)

General

Max. carrying capacity	150 kg ²
Max. blank weight	100 kg
Max. blank dimension (W)	3000 mm
Max. blank dimension (L)	1500 mm
Min. sheet thickness	0.7 mm
Max. loading stack height incl. pallet	900 mm
Max. unloading stack height including pallet (part-dependent)	1500 mm

Floor track

Path length (standard)	8000 m
Traverse path	6450 mm

BendMaster accuracy

Z axis (floor track)	0.5 mm
Overall positioning accuracy of the rotary axes	0.3 mm

Control

Components	TRUMPF
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Vacuum generation

Max. negative pressure	0.7 bar
Max. negative pressure, absolute	0.3 bar
Max. supply volume	500 l/min

Technical Quotation



Connected loads

BendMaster connected load	17.5 kVA
BendMaster weight	784 kg
Floor track of each 2000 mm element	320 kg

Maximum working range

Range (radius)	3400 mm
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¹ Depending on the selected die width and operation cycle.

² Gripper and blank weight

5. Additional information on technical data

TruBend Cell 5170 (S) (KB36)

1700 kN
4250 mm_WRONG
3680 mm_WRONG
615 mm_WRONG
10 mm_WRONG
420 mm_WRONG
180 mm_WRONG
1115 mm_WRONG
1095 mm_WRONG

220 mm/s
10 mm/s
220 mm/s
1000 mm/s
1750 mm/s
330 mm/s
2500 mm/s
1750 mm/s

0.08 mm_WRONG
0.0002 in
0.002 in

445 mm_WRONG
600 mm_WRONG
860 mm_WRONG
250 mm_WRONG

± null None

± null None

132.3 lb
88.2 lb
1000 mm_WRONG
0.7 mm_WRONG
750 mm_WRONG
1000 mm_WRONG

6000 mm_WRONG
4550 mm_WRONG

Technical Quotation



0.5 mm_WRONG

0.2 mm_WRONG

2910 mm_WRONG

Intel Core i5, 2,7 GHz, Dual-Core-Prozessor;
Windows-Embedded-Standard 7; 21,5"-TFT-
Color-Display mit LED-Backlight; 1920 x 1080
Pixel Displayauflösung (Full HD)

0.7 bar

0.3 bar

500 l/min

17.5 kVA

6 bar $\pm$ 1 bar

3000 mm_WRONG

1366.9 lb

39352.1 lb

705.5 lb

4970 mm_WRONG x 1900 mm_WRONG x 3000
mm_WRONG

weiß = NCS S 0505 R80B, blau = ca. RAL-
Design 250 20 20, schwarz = RAL 9005, silberne
Elemente in RAL 9006

BendMaster (150)

General

Max. blank dimension (W) 118.1 in

Max. blank dimension (L) 59.1 in

Min. sheet thickness 0.03 in

Max. loading stack height incl. pallet 35.4 in

Max. unloading stack height including pallet
(part-dependent) 59.1 in

Floor track

Path length (standard) 8748.8 yd

Traverse path 253.9 in

Technical Quotation



BendMaster accuracy

Z axis (floor track)	0.02 in
Overall positioning accuracy of the rotary axes	0.01 in

Control

Components	TRUMPF
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Vacuum generation

Max. negative pressure	0.7 bar
Max. negative pressure, absolute	0.3 bar
Max. supply volume	500 l/min

Connected loads

BendMaster connected load	17.5 kVA
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Maximum working range

Range (radius)	133.9 in
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Chain conveyor for loading and unloading 5725 x 1460 mm (steel pallet 3250 x 1460 mm)

± ± null None

null None x null None x null None

null None x null None x null None

null None ± null None

null None x null None x null None

null None ± null None

null None x null None

null None x null None

null None x null None x null None

null None x null None

null None x null None

± null None

± null None

± null None

Technical Quotation



null None x null None

null None x null None

null - null / null None

null None / null None

null None x null None

null None x null None

Add. pallet for sheet removal station - rotary table

± ± null None

null None x null None x null None

null None x null None x null None

null None ± null None

null None x null None x null None

null None ± null None

null None x null None

null None x null None

null None x null None x null None

null None x null None

null None x null None

± null None

± null None

± null None

null None x null None

null None x null None

null - null / null None

Technical Quotation



null None / null None

null None x null None

null None x null None